DAY 3

Discounted Cash Flow (DCF) Valuation

FMVA Certification Prep: Financial Modeling and Valuation Bootcamp

Lesson Forge Pro (owner)

DCF valuation is the gold standard for estimating intrinsic business value, used by investment bankers, equity analysts, and corporate strategists worldwide. By converting future cash flows into today's dollars, DCF cuts through accounting noise to reveal what a business is truly worth. This chapter builds every skill tested on the FMVA exam: projecting free cash flows, computing WACC, calculating terminal value, assembling the full DCF model, and critically evaluating results through sensitivity analysis. Work through each section sequentially and you will be able to value any company from scratch using only a spreadsheet and the techniques here.

Time Value of Money and the DCF Discount Mechanism

Every valuation model in finance rests on a single, elegant idea: a dollar in your hand today is worth more than a dollar promised in the future. Two forces drive this principle. First, opportunity cost — money received today can be invested immediately to earn a return, so waiting means forgoing that growth. Second, risk — the further into the future a cash flow lies, the less certain you are to actually receive it. Discounted cash flow analysis converts future amounts into their present-day equivalents so that cash flows occurring at different points in time can be compared on a common basis.

The Present Value Formula

The mathematical engine of every DCF model is:

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PV = FV / (1 + r)^n
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Here, FV is the nominal cash flow expected at a future date, r is the discount rate expressed as a decimal, and n is the number of periods until the cash flow is received. Dividing by (1 + r)^n — the discount factor — shrinks the future amount back to its present-day equivalent. The larger r is, or the longer n is, the smaller the resulting PV. This inverse relationship is intuitive: riskier or more distant cash flows are worth less today.

What the Discount Rate Actually Represents

The discount rate r is not a single, universal number. It bundles two distinct concepts. The first is the time cost of money — roughly the risk-free rate you could earn on a safe government bond. The second is a risk premium that compensates investors for the possibility that the projected cash flows never materialize. In corporate finance, the discount rate is typically the Weighted Average Cost of Capital (WACC), which blends the cost of equity and after-tax cost of debt. A higher-risk business demands a higher r, which mechanically reduces the PV of its cash flows and therefore its estimated value.

Compounding Periods Matter

The standard formula assumes annual compounding, but real-world instruments often compound more frequently. For semi-annual compounding, the formula becomes PV = FV / (1 + r/2)^(2n), effectively doubling the number of periods while halving the periodic rate. Continuous compounding — the mathematical limit — uses PV = FV × e^(" rn). Each convention produces a slightly different present value from the same nominal inputs, so always confirm which compounding basis applies before discounting.

From Single Cash Flows to NPV

When a project generates multiple cash flows across several periods, you discount each one individually and then sum the results. Net Present Value (NPV) includes both inflows and outflows:

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NPV = :2 ² 4e÷B ò ƒ  ² "•çB Ð
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A positive NPV means the investment generates more value than it costs in present-value terms — it creates wealth. A negative NPV signals destruction of value. This makes NPV the gold-standard decision rule in capital budgeting.

Worked Example: Discounting a $500,000 Cash Flow

Scenario: A real estate project is expected to generate a single cash flow of $500,000 at the end of Year 3. The appropriate annual discount rate, reflecting both the time value of money and project risk, is 10%. What is the present value of this cash flow?

Step 1 — Identify the inputs.

- FV = $500,000
- r = 0.10
- n = 3

Step 2 — Calculate the discount factor.

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(1 + 0.10)^3 = 1.10^3 = 1.331
```

Step 3 — Apply the present value formula.

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PV = $500,000 / 1.331
```

Step 4 — Compute the result.

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PV = $375,657
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Interpretation: Receiving $500,000 three years from now is economically equivalent to receiving $375,657 today, given a 10% discount rate. The $124,343 difference is the cost of waiting and bearing risk.

Solving in Reverse: Finding the Yield

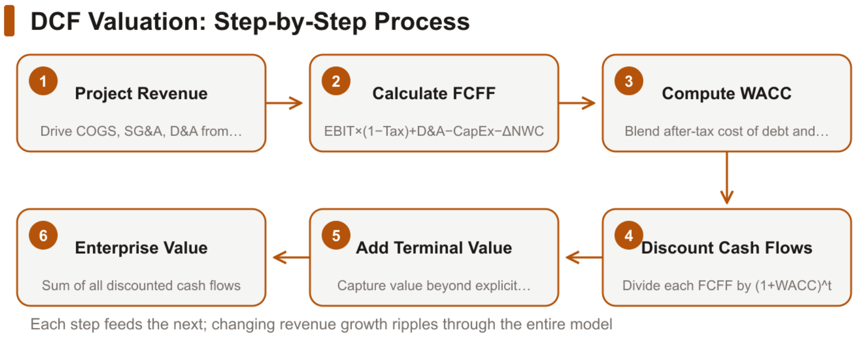
The same formula can be rearranged to solve for r when PV and FV are both known:

```
r = (FV / PV)^(1/n) "  1
```

This is the yield of a single cash flow, and when applied across an entire stream of cash flows it becomes the Internal Rate of Return (IRR) — the discount rate that sets NPV exactly to zero. IRR is widely used to compare projects against a hurdle rate.

> Common mistake: Candidates frequently forget to raise (1 + r) to the power of n and instead simply multiply by (1 + r × n). That is simple interest, not compound discounting, and it will overstate PV for any n greater than 1. Always use the exponent form.

Projecting Free Cash Flow to the Firm (FCFF)



Each step feeds the next; changing revenue growth ripples through the entire model

Free cash flow to the firm (FCFF) is the cash a business generates for all capital providers — equity holders and debt holders alike — after funding operations and reinvestment. Because it is calculated before financing costs, it is the natural output to discount at the weighted average cost of capital (WACC). Mastering a clean, driver-based FCFF model is one of the most tested skills in valuation practice.

The Core Formula

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FCFF = EBIT × (1 " Tax Rate) + D&A " CapEx " 9Dåt0

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Starting from EBIT rather than net income is deliberate. Net income already reflects interest expense, which depends on how the firm is financed. EBIT strips that away, making FCFF capital-structure-neutral and directly comparable across companies with different debt levels.

Linking Revenue to Every Line Item

A well-built model flows from a single top-line assumption. Revenue growth drives everything downstream:

- COGS is modeled as a percentage of revenue (e.g., 50%), producing gross profit.
- SG&A is similarly expressed as a percentage (e.g., 15%), giving EBITDA margin.
- D&A can be set as a percentage of revenue or tied to the opening PP&E balance — either approach works, but be consistent.

This structure means that if you change the revenue growth rate, every margin-based item updates automatically, which is both efficient and auditable.

CapEx: Maintenance vs. Growth

CapEx has two economic purposes. Maintenance CapEx keeps existing assets operational and is often approximated as equal to D&A over the long run. Growth CapEx funds new capacity to support incremental revenue. When possible, separate them: growth CapEx should scale with revenue growth, while maintenance CapEx is more stable. In practice, many models use a single CapEx-to-revenue ratio anchored to historical data and industry benchmarks.

Modeling Net Working Capital Explicitly

Changes in net working capital (NWC) represent cash tied up in — or released from — the operating cycle. Model the three key components using days metrics:

- Days Sales Outstanding (DSO): Receivables = (DSO / 365) × Revenue
- Days Inventory Outstanding (DIO): Inventory = (DIO / 365) × COGS
- Days Payable Outstanding (DPO): Payables = (DPO / 365) × COGS

NWC = Receivables + Inventory " Payables. An increase in NWC is a cash outflow (9Dåt2 —2 7V'G acted); a decrease releases cash.

Step-by-Step Procedure

- Set the base-year revenue and apply the growth rate to project Years 1–5.
- Apply margin assumptions to derive EBITDA, then subtract D&A to reach EBIT.
- Multiply EBIT by (1 " tax rate) to get NOPAT (net operating profit after tax).
- Add back D&A (non-cash charge already deducted from EBIT).
- Subtract total CapEx.
- Calculate NWC for each year using DSO, DIO, and DPO; compute 9Dåt2 •ear-over-year.
- Subtract 9Dåt2 Fò ' ive at FCFF.
- Sanity-check each year against industry benchmarks and the firm's own historical ratios.

Worked Example: Five-Year FCFF Schedule

Assumptions: Base-year revenue = $200M; revenue growth = 8% per year; EBITDA margin = 35%; D&A = $14M (constant, ~7% of base revenue); CapEx = $20M per year; tax rate = 25%; DSO = 45 days, DIO = 30 days, DPO = 40 days; COGS = 50% of revenue.

	Yr 0 (Base)	Yr 1	Yr 2	Yr 3	Yr 4	Yr 5
Revenue ($M)	200.0	216.0	233.3	251.9	272.1	293.9
EBITDA (35%)	70.0	75.6	81.7	88.2	95.2	102.9
D&A	14.0	14.0	14.0	14.0	14.0	14.0
EBIT	56.0	61.6	67.7	74.2	81.2	88.9
NOPAT (×0.75)	42.0	46.2	50.8	55.6	60.9	66.7
+ D&A		14.0	14.0	14.0	14.0	14.0
" CapEx		(20.0)	(20.0)	(20.0)	(20.0)	(20.0)
Receivables	24.7	26.7	28.8	31.1	33.6	36.2
Inventory	8.2	8.9	9.6	10.4	11.2	12.1
Payables	(10.9)	(11.8)	(12.8)	(13.8)	(14.9)	(16.1)
NWC	22.0	23.8	25.6	27.7	29.9	32.2
" 9Dåt2 Â Â ƒ ã,' Â ƒ ã,' Â ƒ"ã ' Â ƒ"ã"' Â ƒ"ã2' À
FCFF		38.4	43.0	47.5	52.7	58.4

Year 1 detail: NOPAT $46.2M + D&A $14.0M " CapEx $20.0M " 9Dåt2 C ã„Ò Ò C3,ãDÒà

> Common mistake: Analysts frequently subtract total CapEx but forget to add back D&A, effectively double-counting the depreciation charge. Remember: D&A was already deducted to reach EBIT, so it must be added back before subtracting actual cash CapEx.

Sanity-Checking the Output

Compare your implied CapEx-to-revenue ratio (~9% here) and NWC-to-revenue ratio (~11%) against industry peers. If your manufacturing peers average 7% CapEx intensity, investigate whether your $20M assumption is realistic or whether growth CapEx is understated. Historical company trends — particularly the relationship between revenue growth and NWC build — are equally important anchors. A model that passes the arithmetic test but fails the benchmark test is still a flawed model.

Calculating WACC as the DCF Discount Rate

The discount rate you plug into a DCF model is not a guess — it is a carefully constructed weighted average of what the firm's capital providers actually require. That rate is the Weighted Average Cost of Capital (WACC), and every component must be grounded in observable market data.

The Core Formula

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WACC = (E/V) × Ke + (D/V) × Kd × (1 " t)

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Here, E is the market value of equity, D is the market value of debt, V = E + D is total firm value, Ke is the cost of equity, Kd is the pre-tax cost of debt, and t is the marginal corporate tax rate. The weights must reflect market values, not book values, because you are pricing risk as the market sees it today, not as the accountant recorded it years ago.

Estimating the Cost of Equity via CAPM

The Capital Asset Pricing Model gives you a theoretically grounded Ke:

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Ke = Rf + ;" r …&Ò !" &b•

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Each input deserves careful attention.

Risk-free rate (Rf): Use the yield on a long-term government bond whose maturity matches your forecast horizon. For a ten-year DCF, the ten-year Treasury (or equivalent sovereign bond) is standard. Avoid short-term T-bill rates — they introduce reinvestment risk that distorts the discount rate.

Equity risk premium (ERP = Rm " Rf): This represents the extra return investors demand for holding equities over risk-free assets. For developed markets, practitioners typically apply 4–6%, anchored by long-run historical data and survey-based estimates (Damodaran updates these annually). For emerging markets, add a country risk premium — often 2–5% depending on sovereign credit spreads — to reflect political and macroeconomic uncertainty.

Beta (;"“ Beta captures systematic, non-diversifiable risk. Because a target firm may be private or have an atypical capital structure, the best practice is to collect betas from publicly traded peers, unlever each one to strip out the peers' financing decisions, average the unlevered betas, and then re-lever using the target's own capital structure:

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;%VæÆPvered = ;$ÆPvered / [1 + (1 " t) × (D/E)]

;%&RÖÆPvered = ;%VæÆPvered × [1 + (1 " t) × (D/E)target]

`

This ensures your beta reflects the business risk of the industry at the financing mix you actually intend to use.

Estimating the After-Tax Cost of Debt

Kd is the pre-tax yield to maturity on the firm's existing bonds or, if those are illiquid, on comparable publicly traded debt with similar credit ratings and maturities. Because interest payments are tax-deductible, the government effectively subsidizes debt financing. The after-tax cost is simply:

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After-tax Kd = Kd × (1 " t)

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Always use the marginal tax rate — the rate applied to the next dollar of income — not the effective rate reported in the financial statements.

Worked Example: Step by Step

A mid-sized industrial manufacturer has the following profile:

Input	Value
Market value of equity (E)	$600 million
Market value of debt (D)	$400 million
Total firm value (V)	$1,000 million
Cost of equity (Ke)	11.2%
Pre-tax cost of debt (Kd)	5.0%
Marginal tax rate (t)	25%

Step 1 — Calculate market-value weights.

- E/V = 600 / 1,000 = 0.60
- D/V = 400 / 1,000 = 0.40

Step 2 — Tax-effect the cost of debt.

- After-tax Kd = 5.0% × (1 " 0.25) = 3.75%

Step 3 — Apply the WACC formula.

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WACC = (0.60 × 11.2%) + (0.40 × 3.75%)

= 6.72% + 1.50%

= 8.22% "H 8.2%

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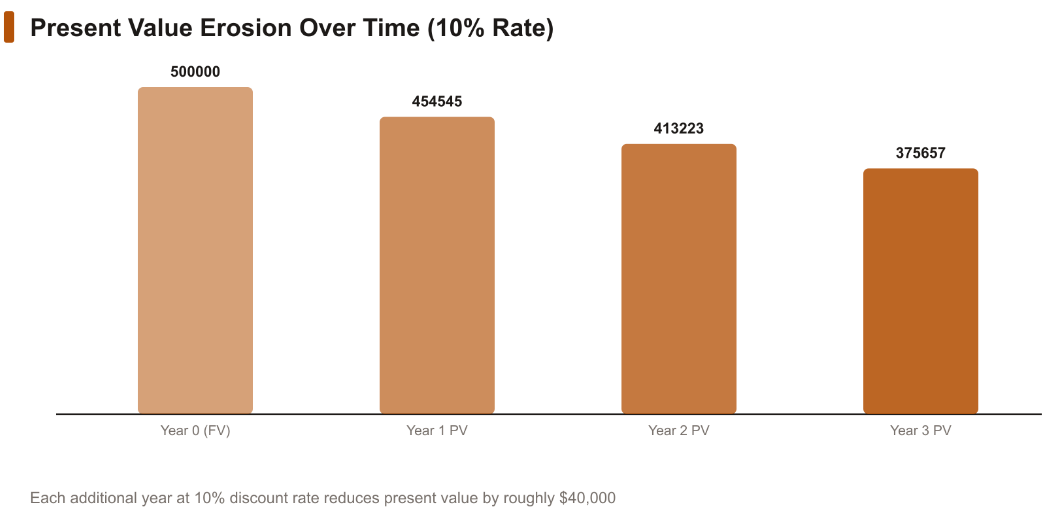
The firm's WACC is 8.2%. Every projected free cash flow in the DCF model will be discounted at this rate to arrive at present value.

Step-by-Step Procedure

- Determine market values of equity and debt; compute E/V and D/V weights.
- Select Rf from the government bond matching your forecast horizon.
- Choose an ERP appropriate for the market (4–6% developed; adjust upward for emerging markets).
- Unlever peer betas, average them, re-lever to the target capital structure.
- Compute Ke = Rf + ;" r U%.
- Identify Kd as the yield to maturity on comparable debt.
- Tax-effect Kd: multiply by (1 " t).
- Combine: WACC = (E/V) × Ke + (D/V) × Kd × (1 " t).

> Common mistake: Using book-value weights instead of market-value weights. Book equity can be dramatically lower than market equity for profitable firms, which overstates the debt weight and artificially depresses WACC — leading to inflated valuations. Always anchor your weights to current market prices.

Terminal Value: Gordon Growth Model and Exit Multiple Methods



Each additional year at 10% discount rate reduces present value by roughly $40,000

Terminal value (TV) is the single largest driver of enterprise value in most DCF models, routinely accounting for 60–80% of total value. Because a small change in your perpetuity growth rate or exit multiple can swing the result by hundreds of millions of dollars, understanding how to build and cross-check terminal value is one of the most consequential skills in financial analysis.

Why Terminal Value Exists

Explicit forecast periods typically run five to ten years—long enough to model near-term business dynamics, but not long enough to capture the full life of a going concern. Terminal value captures everything beyond that horizon in a single, tractable number. Two methods dominate practice: the Gordon Growth Model (also called the perpetuity growth method) and the Exit Multiple Method.

Gordon Growth Model

The Gordon Growth Model treats the business as a perpetuity growing at a constant rate after the final forecast year. The formula is:

```
TV = FCFFn+1 / (WACC "  g)
```

Here, FCFFn+1 is free cash flow to the firm in the first year beyond the explicit forecast period, WACC is the weighted average cost of capital, and g is the long-run perpetuity growth rate.

The most important discipline when applying this formula is constraining g. Because the model assumes growth continues forever, setting g above the long-run nominal GDP growth rate implies the company will eventually become larger than the entire economy—a logical impossibility. In practice, analysts limit g to roughly 2–3%, consistent with long-run nominal GDP in developed markets. For mature, slow-growing industries, 1.5–2% is more defensible.

Exit Multiple Method

Rather than assuming a perpetuity, the Exit Multiple Method anchors terminal value to observable market pricing. You apply a valuation multiple—most commonly EV/EBITDA—derived from comparable public companies or precedent transactions to the final forecast year's EBITDA:

```
TV = EBITDAn × EV/EBITDA Multiple
```

The selected multiple should reflect where comparable companies trade today, with judgment applied for any expected changes in industry structure or growth prospects by the end of the forecast period.

Discounting Terminal Value to the Present

Regardless of which method you use, terminal value is a future amount and must be discounted back to today:

```
PV(TV) = TV / (1 + WACC)^n
```

where n is the length of the explicit forecast period in years.

Fully Worked Example

Given information:

- Year-5 FCFF (FCFFn): $85 million
- WACC: 8.2%
- Perpetuity growth rate (g): 2.5%
- Year-5 EBITDA: $112 million
- Selected EV/EBITDA exit multiple: 7×
- Explicit forecast period: 5 years

Step 1 — Calculate FCFFn+1 for the Gordon Growth Model.

FCFFn+1 = $85M × (1 + 0.025) = $87.125 million

Step 2 — Apply the Gordon Growth Model.

```
TV = $87.125M / (0.082 "  0.025) = $87.125M / 0.057
```

TV = $1,528.5 million

Step 3 — Discount the Gordon Growth TV to present value.

```
PV(TV) = $1,528.5M / (1.082)^5
```

(1.082)^5 = 1.4834 (calculated as 1.082 × 1.082 × 1.082 × 1.082 × 1.082)

PV(TV) = $1,528.5M / 1.4834 = $1,030.4 million

Step 4 — Apply the Exit Multiple Method.

```
TV = $112M × 7 = $784.0 million
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Step 5 — Discount the Exit Multiple TV to present value.

PV(TV) = $784.0M / 1.4834 = $528.5 million

Step 6 — Cross-check the two methods.

The Gordon Growth Model yields a PV of $1,030.4M versus the Exit Multiple's $528.5M—a divergence of nearly 95%. This is a significant red flag. In practice, you would revisit your assumptions: Is g = 2.5% too aggressive for this business? Does a 7× EBITDA multiple understate the peer group? Narrowing the spread—ideally to within 10–20%—gives you confidence that your terminal value is grounded in reality.

> Common mistake: Analysts frequently forget to grow FCFFn by (1 + g) before dividing in the Gordon Growth formula. Using the Year-5 FCFF directly—rather than the Year-6 amount—understates terminal value and introduces a systematic error in every deal model.

Procedure Summary

- Confirm your explicit forecast period ends at year n.
- Grow final-year FCFF by (1 + g) to obtain FCFFn+1.
- Apply TV = FCFFn+1 / (WACC " g); verify g "d long-run nominal GDP growth.

- Independently compute TV = EBITDAn × comparable multiple.
- Discount both terminal values using PV(TV) = TV / (1 + WACC)^n.
- Compare results; investigate any divergence exceeding ~15–20% before finalizing.

Assembling the Full DCF Model and Deriving Equity Value

Pulling a DCF model together is the moment all the individual pieces—projected free cash flows, a carefully chosen discount rate, and a defensible terminal value—finally speak as one coherent answer. The goal of this section is to show you exactly how those pieces connect, how to translate enterprise value into a per-share intrinsic value, and how to build the model so it remains clean and testable long after you first construct it.

From Projected Cash Flows to Enterprise Value

Enterprise value in a DCF framework is simply the present value of everything the business will ever generate for its capital providers, expressed in today's dollars:

```
Enterprise Value = PV of Explicit-Period FCFFs + PV of Terminal Value
```

The explicit period (commonly five to ten years) captures the years you can forecast with reasonable confidence. The terminal value captures all remaining cash flows beyond that horizon, typically using either a Gordon Growth Model or an exit-multiple approach. Both components are discounted at the weighted average cost of capital (WACC), which reflects the blended required return of debt and equity holders.

Bridging to Equity Value

Enterprise value belongs to all capital providers—debt holders, equity holders, and sometimes preferred shareholders. To isolate what belongs to common equity, you subtract net debt and add back any non-operating assets (such as excess cash held in a subsidiary or an equity investment not already reflected in the FCFFs):

```
Equity Value = Enterprise Value "  Net Debt
```

where Net Debt = Total Debt " Cash and Cash Equivalents. Non-operating assets are added because they have value but were excluded from the operating cash flow projections.

Per-Share Intrinsic Value

Once equity value is in hand, divide by diluted shares outstanding—not basic shares. Diluted shares include the effect of in-the-money stock options (using the treasury stock method) and any convertible securities that would increase the share count upon conversion. Using basic shares overstates per-share value.

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Equity Value per Share = Equity Value ÷ Diluted Shares Outstanding
```

The Mid-Year Convention

Standard end-of-year discounting assumes every dollar of cash flow arrives on December 31. In reality, a healthy operating business generates cash throughout the year. The mid-year convention corrects for this by treating each year's cash flow as if it is received at the midpoint—period 0.5, 1.5, 2.5, and so on. The practical effect is a slightly higher present value, because cash is assumed to arrive, on average, six months earlier than the end-of-year assumption implies. For Year 1, the discount factor becomes 1 / (1 + WACC)^0.5 rather than 1 / (1 + WACC)^1.

Fully Worked Numerical Example

Given information:

- Sum of discounted Year 1–5 FCFFs: $180 million
- PV of terminal value: $620 million
- Total debt: $200 million; Cash: $80 million !' Net Debt = $120 million
- Diluted shares outstanding: 34 million

Step 1 — Calculate Enterprise Value

$180M + $620M = $800M Enterprise Value

Step 2 — Subtract Net Debt

Net Debt = $200M " $80M = $120M

$800M " $120M = $680M Equity Value

Step 3 — Divide by Diluted Shares

$680M ÷ 34M shares = $20.00 per share intrinsic value

Model Architecture Best Practices

A well-structured DCF model follows a deliberate tab sequence that mirrors the analytical logic:

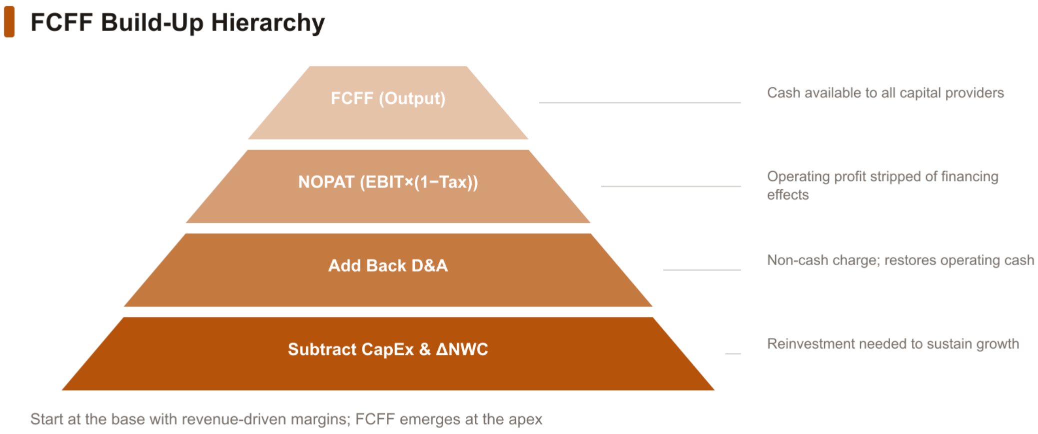
- Assumptions tab — all drivers (revenue growth, margins, capex %, WACC, terminal growth rate) live here as hard-coded inputs.
- Income statement — revenue through NOPAT, pulling from assumptions.
- Balance sheet — working capital, PP&E, and debt schedules.
- Cash flow statement — derives FCFF from operating and investing activity.
- DCF output tab — discounts FCFFs, computes terminal value, and performs the enterprise-to-equity bridge.

The single most important structural rule is to hard-code every input in one place and reference it everywhere else. This discipline means that when you run a sensitivity table varying WACC from 8% to 12% or terminal growth from 2% to 4%, every downstream cell updates automatically without hunting for buried constants.

> Common mistake: Analysts sometimes subtract gross debt instead of net debt, forgetting to add back cash. If a company holds $80M in cash, that cash is available to pay down debt or return to shareholders—ignoring it understates equity value. Always confirm whether "cash" in the balance sheet includes restricted cash, which should not be netted against debt.

With these mechanics in place, your DCF model produces a transparent, auditable intrinsic value estimate that can be stress-tested systematically—exactly what examiners and practitioners expect.

Sensitivity Analysis and Critical Evaluation of DCF Outputs



Start at the base with revenue-driven margins; FCFF emerges at the apex

Sensitivity analysis transforms a DCF model from a single fragile number into a defensible range of values — and that shift from point estimate to range is exactly what separates competent analysts from overconfident ones.

Why a Single DCF Output Is Never Enough

Every DCF rests on assumptions about future free cash flows, the weighted average cost of capital (WACC), and the terminal growth rate (g). Each assumption carries uncertainty, and those uncertainties compound. Presenting one equity value per share as though it were a fact invites serious exam and real-world criticism. Sensitivity analysis forces you to quantify how much the conclusion changes when the inputs move — and the answer is almost always "more than you expected."

Two-Variable Data Tables: WACC vs. Terminal Growth Rate

In Excel, a two-variable data table lets you hold all other inputs constant while simultaneously varying two drivers across a grid. The most informative combination is WACC on one axis and the terminal growth rate on the other, because these two parameters overwhelmingly determine terminal value (TV), which typically represents 60–80 % of total enterprise value in a five-year DCF.

Worked Example — 5 × 5 Sensitivity Table

Assume a company with projected Year 5 free cash flow of $2.10 per share, a base WACC of 8 %, a base terminal growth rate of 2.5 %, and present value of the explicit forecast period equal to $6.50 per share (already discounted). Net debt per share is $2.00.

Terminal value per share (Gordon Growth Model):

```
TV per share = FCF … × (1 + g) / (WACC "  g)
```

At base case: TV = 2.10 × 1.025 / (0.08 " 0.025) = 2.1525 / 0.055 = $39.14 per share (Year 5 value)

Discounted to today at 8 % over 5 years: 39.14 / (1.08) u = 39.14 / 1.4693 = $26.64

Equity value per share = PV of forecast period + PV of TV " net debt = 6.50 + 26.64 " 2.00 = $31.14 (base case)

Repeating this calculation across all 25 WACC/g combinations produces the table below. Values are equity value per share, rounded to the nearest dollar.

	g = 1.5%	g = 2.0%	g = 2.5%	g = 3.0%	g = 3.5%
WACC = 7.0%	$21	$23	$26	$28	—
WACC = 7.5%	$19	$21	$23	$25	$28
WACC = 8.0%	$17	$19	$21	$23	$25
WACC = 8.5%	$16	$17	$19	$21	$23
WACC = 9.0%	$14	$15	$17	$18	$20

The range — $14 to $28 — spans a factor of two from a mere 200 basis-point shift in WACC and 200 basis points in g. That is the central lesson of sensitivity analysis.

The Key Formula: Terminal Value Sensitivity

The approximate percentage change in terminal value for a small change in g is:

```
% Change in TV "H " (1 / (WACC "  g)) × 9Fp
```

At base case (WACC = 8 %, g = 2.5 %), a 0.5 % increase in g produces: " (1 / 0.055) × (" 0.005) "H +9.1 % change in TV. That single half-point move in a growth assumption shifts terminal value by nearly a tenth — before any compounding with forecast-period errors.

Tornado Charts and Input Ranking

A tornado chart ranks every DCF input by its impact on equity value, with the widest bars at the top. In virtually every corporate DCF, WACC and terminal growth rate occupy the top two positions. Revenue growth rate in the explicit period typically ranks third, followed by operating margin assumptions and capital expenditure intensity. This ranking tells you where to spend diligence time and where to anchor management conversations.

> Common mistake: Analysts often spend hours refining Year 3 capex assumptions while leaving WACC and g at round-number defaults. The tornado chart makes clear that this effort is misallocated — the two dominant drivers deserve the most rigorous justification.

Strengths and Weaknesses of DCF

DCF's core strength is that it is grounded in economic fundamentals: the intrinsic value of any asset equals the present value of the cash it generates. It is immune to the comparable-selection bias that plagues multiples analysis, where cherry-picking peers can shift a valuation by 30 % or more.

Its core weakness is the garbage-in-garbage-out problem illustrated above. Small, defensible-sounding changes in WACC or g produce enormous value swings. The model rewards precision in inputs it cannot actually deliver.

Triangulation and Scenario Analysis

Best practice triangulates the DCF output with comparable company analysis (trading multiples) and precedent transaction analysis (deal multiples). If all three methods cluster around $20–$24 per share, confidence rises. If DCF says $31 and comps say $18, the gap demands explanation — not averaging.

Scenario analysis (base, bull, bear) communicates uncertainty more honestly than any single estimate. Assign each scenario a narrative (e.g., bear = margin compression plus rate rise) and a probability weight. The probability-weighted value becomes your anchor; the range becomes your communication tool with clients, boards, and examiners.

Glossary

- Discounted Cash Flow (DCF) — Valuation method converting future cash flows to present value using a discount rate.
- Free Cash Flow to the Firm (FCFF) — Operating cash flow after tax and reinvestment, available to all capital providers.
- WACC — Blended after-tax cost of debt and equity weighted by market-value capital structure proportions.
- Terminal Value — Present value of all cash flows beyond the explicit forecast horizon, often as a perpetuity.
- Beta (;"' r ÖV 7W&R öb 7Fö0k's systematic risk relative to the overall market.
- Equity Risk Premium (ERP) — Excess return investors demand for holding equities over the risk-free rate.
- Net Present Value (NPV) — Sum of all discounted cash inflows and outflows; positive NPV indicates value creation.
- Perpetuity Growth Rate (g) — Assumed constant long-run growth rate of terminal cash flows in the Gordon Growth Model.